

```

clc;

clear;

close all;


% EX NO: 2

% Chebyshev Type-I IIR Low-Pass Filter

% Date:


% Filter Specifications

Wp = 0.40; % Passband Edge (Normalized)

Ws = 0.55; % Stopband Edge (Normalized)

Rp = 1; % Passband Ripple (dB)

As = 40; % Stopband Attenuation (dB)


% Find Minimum Filter Order

[N, Wn] = cheb1ord(Wp, Ws, Rp, As);


% Design Chebyshev Type-I Low-Pass Filter

[b, a] = cheby1(N, Rp, Wn, 'low');


% Display Results

fprintf('Filter Order (N) = %d\n', N);

fprintf('Normalized Cutoff Frequency (Wn) = %.4f\n', Wn);


% Stability Check

if all(abs(roots(a)) < 1)

    disp('System is STABLE');

else

```

```

        disp('System is UNSTABLE');
end

% -----
% Magnitude and Phase Response
% -----

figure;
freqz(b, a, 1024);
title('Magnitude and Phase Response');

% -----
% Impulse Response
% -----

figure;
impz(b, a);
grid on;
title('Impulse Response');

% -----
% Group Delay
% -----

figure;
grpdelay(b, a, 1024);
grid on;
title('Group Delay');

% -----
% Pole-Zero Plot

```

```
% -----
```

```
figure;
```

```
zplane(b, a);
```

```
grid on;
```

```
title('Pole-Zero Plot');
```



